

A Scatter Plot with Linear Regression analysis is a quick way to visualize trends in continuous data variables.

Step 1: Select samples

Drag and drop terms of interest into the Subset boxes on the **Comparison** tab. Include all terms to be used in the analysis.

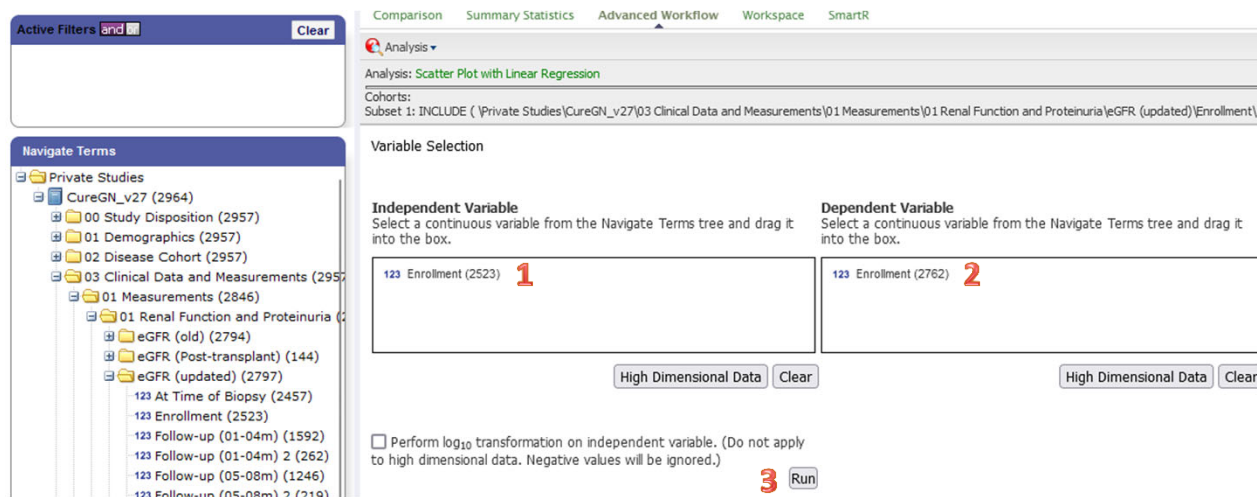
Click on the **Summary Statistics** tab to review the number of samples that will be included in the analysis. Confirm that the set of samples available is sufficient for running a meaningful analysis.

Step 2: Set up analysis

Click on the **Advanced Workflow** tab and select "Scatter Plot with Linear Regression" from the Analysis dropdown list.

Drag terms from the Navigate Terms tree into the Independent Variable (1) and Dependent Variable (2) boxes. Both boxes should contain continuous variables (e.g. EGFR at Enrollment visit and BMI at Enrollment visit).

Click the "Run" button (3). A popup box will show a progress bar as the data is compiled and analyzed by tranSMART. Some analyses may take several minutes to run.



The screenshot shows the tranSMART interface with the 'Advanced Workflow' tab selected. The 'Analysis' dropdown is set to 'Scatter Plot with Linear Regression'. The 'Cohorts' section shows a subset of 'INCLUDE' with a list of terms. The 'Variable Selection' section has two boxes: 'Independent Variable' (1) and 'Dependent Variable' (2). Both boxes contain '123 Enrollment (2523)'. Below these boxes are buttons for 'High Dimensional Data' and 'Clear'. A checkbox for 'Perform log₁₀ transformation on independent variable' is unchecked. A red '3' and a 'Run' button are at the bottom right.

Step 3: Review results

Results will appear underneath the variable selection boxes (4). The Independent variable will appear along the X-axis and the Dependent variable will be the Y-axis.

Scroll below the box plot to see numerical Linear Regression results (5).

5

Linear Regression Result

Number of Subjects	2461
Intercept	31.3
Slope	-0.0561
r-squared	0.0636
adjusted r-squared	0.0632
p-value	5.42e-37

4

